

- 5 (a) In 1965, Richard Feynman hypothesised that electrons could be used in the double slit experiment to demonstrate wave particle duality. This experiment has since been conducted and verified by physicists.

A simplified set-up of the experiment is shown in Fig. 5.1.

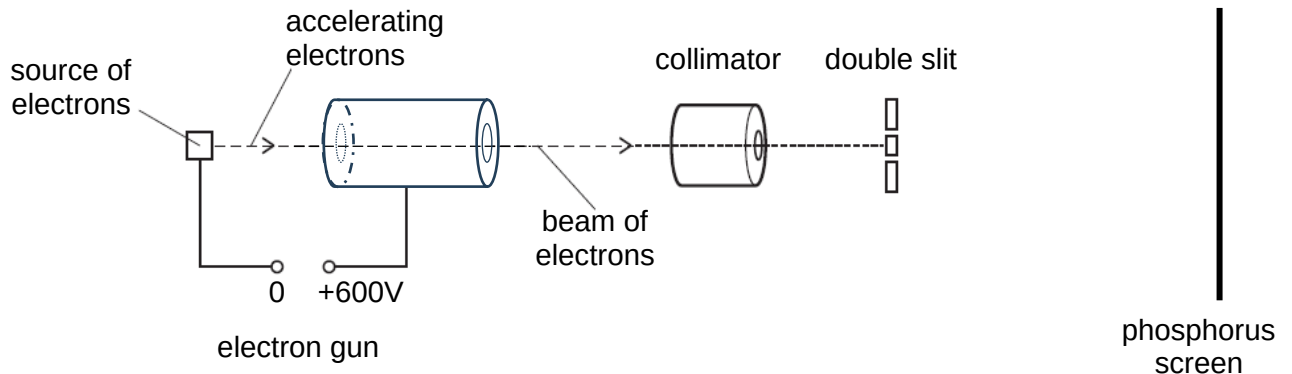


Fig. 5.1

An electron gun accelerates electrons in vacuum through a potential difference of 600 V. The collimator narrows the beam of electrons which then passes through the double slit.

The double slit only allows electrons to pass through one at a time, with each electron detected as a single bright spot on the phosphorus screen. The bright spots cumulate with time, showing a pattern formed on the screen as shown in Fig. 5.2.

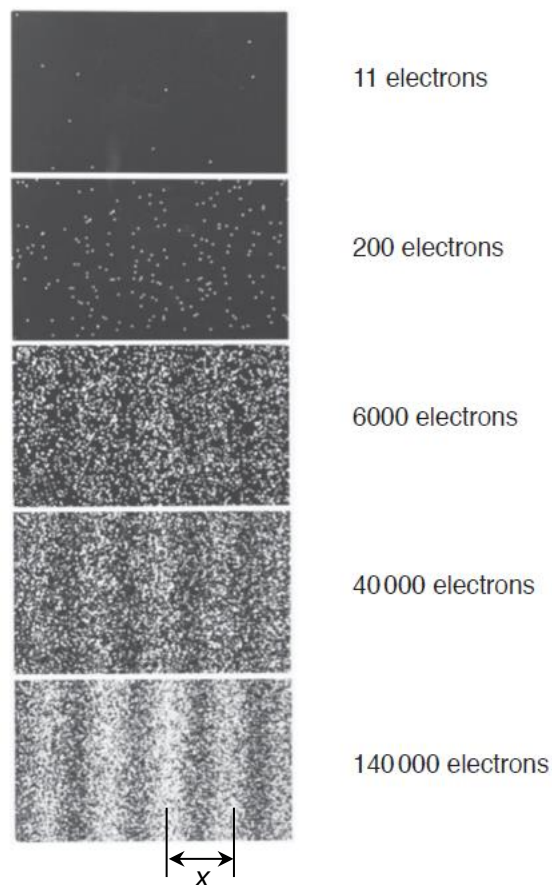


Fig. 5.2 (not to scale)

- (i) Explain how the images in Fig. 5.2 show that electrons exhibit both particle-like and wave-like properties.

Particle property: Each electron registers as a single point/dot on the screen [B1]

Wave property: Regular maxima and minima (or bright and dark regions) indicates an interference pattern associated with waves [B1]

.....[2]

- (ii) Calculate the de Broglie wavelength of the electrons reaching the double slit.

$$\begin{aligned}
 qV &= \frac{1}{2}mv^2 & qV &= \frac{p^2}{2m} \\
 v &= \sqrt{\frac{2qV}{m}} & p &= \sqrt{2mqV} \\
 &= \sqrt{\frac{2(1.60 \times 10^{-19})(600)}{9.11 \times 10^{-31}}} & &= \sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(600)} \\
 &= 1.452 \times 10^7 \text{ m s}^{-1} \text{ [C1]} & &= 1.322 \times 10^{-23} \text{ N s [C1]} \\
 \lambda &= \frac{h}{mv} & \lambda &= \frac{h}{p} \\
 &= \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(1.452 \times 10^7)} \text{ [C1]} & &= \frac{6.63 \times 10^{-34}}{1.322 \times 10^{-23}} \text{ [C1]} \\
 &= 5.01 \times 10^{-11} \text{ m [A1]} & &= 5.02 \times 10^{-11} \text{ m [A1]} \\
 \text{wavelength} &= \dots\dots\dots 5.01 \times 10^{-11} \dots\dots\dots \text{ m [3]}
 \end{aligned}$$

- (iii) The double slit consists of two 50 nm wide slits with a separation of 280 nm. The distance from the double slit to the screen is 1.2 m.

Calculate the expected value of the centre-to-centre distance x shown in the last image of Fig. 5.2.

$$\begin{aligned}
 x &= \frac{\lambda D}{a} \\
 &= \frac{(5.01 \times 10^{-11})(1.2)}{280 \times 10^{-9}} \text{ [C1] ecf their de Broglie } \lambda \\
 &= 2.15 \times 10^{-4} \text{ m [A1]} \\
 x &= \dots\dots\dots 2.15 \times 10^{-4} \dots\dots\dots \text{ m [2]}
 \end{aligned}$$

- (iv) In order for clear patterns to be formed, suggest why it is important for the electrons reaching the double slit to have velocities very close to a single value.

Since velocity affects the de Broglie wavelength ($\lambda = \frac{h}{mv}$) of the electron, a single de Broglie wavelength is needed to get a clear/observable interference pattern (so distance x can be similar) [B1]

.....[1]

- (b) In 1927, Werner Heisenberg proposed that we can only be clear about what is meant by the position of an object if we can specify experiments by which its position can be measured.

Heisenberg realised that a microscope using visible light would be useless for making a precise measurement of the position of something as small as an electron. He thought of using a 'gamma ray microscope', where gamma rays bounce off the electron and into a measuring device which can be used to determine the location of the electron.

- (i) Explain why scattering light or gamma rays bounce off an electron causes the electron to change its momentum.

Photons have momentum [B1]

When photons bounce off/scattered by the electron, a collision occurs and momentum is transferred to the electron (by the principle of conservation of momentum), causing the electron's momentum to change [B1].....[2]

- (ii) By reference to how the momentum change of an electron depends on the wavelength of the electromagnetic radiation used to observe it, explain why using gamma rays could allow a more precise measurement of position to be made.

(Since $p = \frac{h}{\lambda}$) Shorter wavelength of EM radiation leads to greater momentum transfer

and change imparted to the electron [B1]

(By $\Delta x \Delta p \geq h$) so there is greater uncertainty in momentum Δp so uncertainty in position

Δx is minimised (allowing for more precise measurement of position to be made) [B1].....[2]

[Total: 12]

6 (a) (i) Define *decay constant*